

Data Structures 1 - 25 Winter

Assignment 01 (Dry)

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Problem 01: Big-O Simplification

Simplify each expression to **Big-O notation**. Justify your answer using the definition of Big-O and/or its properties.

1. $O(5n^3 + 3n^2 + 100n + 1000) + O(n \cdot \log n)$
2. $O(2^n) \cdot O(n^2) + O(3^n)$
3. $O(\log_2 n) \cdot O(\sqrt{n}) + O(n \cdot \log n)$
4. $O(n!) + O(2^n) \cdot O(n^3)$

1. $O(5n^3 + 3n^2 + 100n + 1000) + O(n \cdot \log n)$

Firstly,

$$O(5n^3 + 3n^2 + 100n + 1000) = O(n^3).$$

Moreover,

$$\lim_{n \rightarrow \infty} \frac{n^3}{n \cdot \log n} = \infty \implies n^3 \gg n \cdot \log n.$$

Therefore,

$$O(5n^3 + 3n^2 + 100n + 1000) + O(n \cdot \log n) = O(n^3) + O(n \cdot \log n) = O(n^3).$$

2. $O(2^n) \cdot O(n^2) + O(3^n)$

Firstly,

$$O(2^n) \cdot O(n^2) = O(2^n \cdot n^2).$$

Moreover,

$$\lim_{n \rightarrow \infty} \frac{3^n}{2^n \cdot n^2} = \lim_{n \rightarrow \infty} \frac{\left(\frac{3}{2}\right)^n}{n^2} = \infty \implies 3^n \gg 2^n \cdot n^2.$$

Therefore,

$$O(2^n) \cdot O(n^2) + O(3^n) = O(2^n \cdot n^2) + O(3^n) = O(3^n).$$

3. $O(\log_2 n) \cdot O(\sqrt{n}) + O(n \cdot \log n)$

Firstly,

$$O(\log_2 n) \cdot O(\sqrt{n}) = O(\sqrt{n} \cdot \log_2 n).$$

Moreover,

$$\lim_{n \rightarrow \infty} \frac{n \cdot \log n}{\sqrt{n} \cdot \log_2 n} = \lim_{n \rightarrow \infty} \sqrt{n} = \infty \implies n \cdot \log n \gg \sqrt{n} \cdot \log_2 n.$$

Therefore,

$$O(\log_2 n) \cdot O(\sqrt{n}) + O(n \cdot \log n) = O(\sqrt{n} \cdot \log_2 n) + O(n \cdot \log n) = O(n \cdot \log n).$$

$$4. O(n!) + O(2^n) \cdot O(n^3)$$

Firstly,

$$O(2^n) \cdot O(n^3) = O(2^n \cdot n^3).$$

Moreover,

$$\lim_{n \rightarrow \infty} \frac{n!}{2^n \cdot n^3} = \infty \implies n! \gg 2^n \cdot n^3.$$

Therefore,

$$O(n!) + O(2^n) \cdot O(n^3) = O(n!) + O(2^n \cdot n^3) = O(n!).$$

Problem 2

(a) State the precondition required for correctness

Pre-condition:

1. array A is not null
2. the range of elements in array A are in [0,9] otherwise the index of B will out of bound

Thus this algorithm only can be used to sort the array whose elements are in [0,9]

(b) Determine the worst-case time complexity. Provide a detailed asymptotic analysis

Overall, the idea of algorithm is:

1. Create an array B with size 10(from 0 to 9), this array B is used to count the times of same digit respect to the index
For example if 4 appears 5 times in array A, then in array B, in the index 4, B[4] = 5
2. Then we create the array A again according to the array B, that is if B[0] = 5, which means digit 0 appears 5 times. Then we modify array A: Let position 0 to position 4 be 0 and so on...

Then let's analyze step by step, assume the length of array A is n .

1. `int B[] = new int[]{0,0,0,0,0,0,0,0,0,0};` is cost c_1 with times 1

Thus this first item of $T(n)$ is $c_1 \times 1$

```
1  for(int i = 0; i < A.length; i++) {
2      int index = A[i];
3      B[index] = B[index] + 1;
4  }
```

This is a loop

`for(int i = 0; i < A.length; i++) {` is cost c_2 with times $n + 1$ since i needs to be n to check condition

`int index = A[i];` is cost c_3 with times n

`B[index] = B[index] + 1;` is cost c_4 with times n

Thus this second item of $T(n)$ is $c_2 \times (n + 1) + c_3 \times n + c_4 \times n$

3. `int k = 0;` is cost c_5 with times 1

Thus this third item of $T(n)$ is $c_5 \times 1$

```

1  for(int i = 0; i <= 9; i++) {
2      int j = 0;
3      while(j < B[i]) {
4          A[k] = i;
5          k = k + 1;
6          j = j + 1;
7      }
8  }

```

This is a loop.

`for(int i = 0; i <= 9; i++)` is cost c_6 with times 11 since i needs to be 10 to check condition

`int j = 0;` is cost c_7 with times 10

`while(j < B[i]) {` is cost c_8 with times $10 \times n$ since it is in the for loop, then 10 times. Also for this while loop, it ranges from 0 to `B[i]`

And we know the elements of `B[i]` is the appearing times of the elements in array A

Thus in the worst case: all element is in `B[i]`, then $\max(B[i]) = n$

`A[k] = i;` is cost c_9 with times n

`k = k + 1;` is cost c_{10} with times n

`j = j + 1;` is cost c_{11} with times n

Thus this fourth item of $T(n)$ is $c_6 \times 11 + c_7 \times 10 + c_8 \times 10 \times n + c_9 \times n + c_{10} \times n + c_{11} \times n$

Thus

$$T(n) = c_1 \times 1 + c_2 \times (n + 1) + c_3 \times n + c_4 \times n + c_5 \times 1 + c_6 \times 11 + c_7 \times 10 + c_8 \times 10 \times n + c_9 \times n + c_{10} \times n + c_{11} \times n$$

$$\text{Then } T(n) = (c_2 + c_3 + c_4 + 10c_8 + c_9 + c_{10} + c_{11})n + (c_1 + c_2 + c_5 + 11c_6 + 10c_7)$$

When n goes to infinity, $T(n) \approx (c_2 + c_3 + c_4 + 10c_8 + c_9 + c_{10} + c_{11})n \leq c'n$ for

$$c' = (c_2 + c_3 + c_4 + 10c_8 + c_9 + c_{10} + c_{11})$$

Thus it is linear time: $O(n)$ in worst case

(c) Prove correctness using the loop invariant

Since there are two loops, then we check the loop invariant twice

1. PC = `{A != null && 0 <= A[p] <= 9 for all p s.t. 0 <= p <= A.length - 1 && B[k] = 0 for all k s.t. 0 <= k <= 9 && i = 0}`

Inv = `{0 <= i <= A.length && B[k] = |\{p : A[p] = k && 0 <= p < i\}| for all 0 <= k <= 9}`

B = `{0 <= i < A.length}`

QC = `{i = A.length && B[k] = |\{p : A[p] = k && 0 <= p < A.length\}| for all 0 <= k <= 9}`

Then we prove correctness by loop invariant theorem

1. Initialization: $Pc \implies Inv$

Suppose we have PC, then `i = 0` and `A != null` $\implies$ `i = 0` and `A.length >= 0` $\implies$ `0 <= i <= A.length`

Since `i = 0`, then `\{p : A[p] = k && 0 <= p < i\}` is empty, then `B[k] = 0`

Since `B[k] = 0 for all k s.t. 0 <= k <= 9 && i = 0` and `0 <= A[p] <= 9 for all p s.t. 0 <= p <= A.length - 1`, then `B[k] = |\{p : A[p] = k && 0 <= p < i\}| for all 0 <= k <= 9`

Thus we prove the invariant holds

2. Maintenance: Assuming Inv && B holds, the execution of the loop body makes Inv true again

Suppose we have Inv and B, NTP Inv is true again

`Inv && B = {0 <= i0 < A.length && B[k] = |\{p : A[p] = k && 0 <= p < i0\}| for all 0 <= k <= 9}`

9}

Let $i = i_0 + 1$, then $0 \leq i_0 + 1 \leq A.length$, then $0 \leq i \leq A.length$

Also since $B[k] = |\{p : A[p] = k \ \&\& \ 0 \leq p < i_0\}|$ for all $0 \leq k \leq 9$, then $B[k] = |\{p : A[p] = k \ \&\& \ 0 \leq p < i_0 + 1\}|$ for all $0 \leq k \leq 9$

Then $B[k] = |\{p : A[p] = k \ \&\& \ 0 \leq p < i\}|$ for all $0 \leq k \leq 9$

Thus Inv is true again

3. Termination: $Inv \ \&\& \ \text{not } B \implies Qc$

$Inv \ \&\& \ \text{not } B = \{0 \leq i \leq A.length \ \&\& \ i \geq A.length \ \&\& \ B[k] = |\{p : A[p] = k \ \&\& \ 0 \leq p < i\}|$
for all $0 \leq k \leq 9\}$

Then $Inv \ \&\& \ \text{not } B = \{i = A.length \ \&\& \ B[k] = |\{p : A[p] = k \ \&\& \ 0 \leq p < A.length\}|$ for all $0 \leq k \leq 9\}$

Thus Qc holds

2. 1. For outer for loop

$PC = \{A \neq \text{null} \ \&\& \ 0 \leq B[v]$ for all v s.t. $0 \leq v \leq 9 \ \&\& \ B[0] + B[1] + \dots + B[9] = A.length$
 $\ \&\& \ k = 0 \ \&\& \ i = 0\}$

$Inv = \{0 \leq i \leq 10 \ \&\& \ k = B[0] + B[1] + \dots + B[i-1] \ \&\& \ |\{p : 0 \leq p < k \ \&\& \ A[p] = v\}| = B[v]$
for all v s.t. $0 \leq v < i \ \&\& \ |\{p : 0 \leq p < k \ \&\& \ A[p] = v\}| = 0$ for all v s.t. $i \leq v \leq 9\}$

$B = \{i \leq 9\}$

$QC = \{k = A.length \ \&\& \ |\{p : 0 \leq p < A.length \ \&\& \ A[p] = v\}| = B[v]$ for all v s.t. $0 \leq v \leq 9$
 $\ \&\& \ \text{isSorted}(A)\}$

Then we prove correctness by loop invariant theorem

1. Initialization: $PC \implies Inv$

Since $i = 0$ and $k = 0$, we have $k = B[0] + \dots + B[-1] = 0$ holds and $|\{p : 0 \leq p < 0 \ \&\& \ A[p] = v\}| = 0$ for all v with $0 \leq v \leq 9$

Thus Inv holds

2. Maintenance: Assuming $Inv \ \&\& \ B1$ holds, the execution of the loop body makes Inv true again

Suppose we have Inv and B, NTP Inv is true again

$Inv \ \&\& \ B = \{0 \leq i_0 \leq 9 \ \&\& \ k = B[0] + B[1] + \dots + B[i_0-1] \ \&\& \ |\{p : 0 \leq p < k \ \&\& \ A[p] = v\}| = B[v]$ for all v s.t. $0 \leq v < i_0 \ \&\& \ |\{p : 0 \leq p < k \ \&\& \ A[p] = v\}| = 0$ for all v s.t. $i_0 \leq v \leq 9\}$

Let $i = i_0 + 1$

Since $0 \leq i_0 \leq 9$, then $0 \leq i \leq 10$

Since $k = B[0] + B[1] + \dots + B[i_0-1]$, $|\{p : 0 \leq p < k \ \&\& \ A[p] = v\}| = B[v]$ for all v s.t. $0 \leq v < i_0$ and $|\{p : 0 \leq p < k \ \&\& \ A[p] = v\}| = 0$ for all v s.t. $i_0 \leq v \leq 9$

By the inner loop's Qc(proved below), we have $k = B[0] + \dots + B[i_0-1] + B[i_0]$ and $|\{p : 0 \leq p < k \ \&\& \ A[p] = i_0\}| = B[i_0]$

Then we have $k = B[0] + B[1] + \dots + B[i-1] \ \&\& \ |\{p : 0 \leq p < k \ \&\& \ A[p] = v\}| = B[v]$ for all v s.t. $0 \leq v < i$ and $|\{p : 0 \leq p < k \ \&\& \ A[p] = v\}| = 0$ for all v s.t. $i \leq v \leq 9$

Thus Inv holds again

3. Termination: $Inv \ \&\& \ \text{not } B1 \implies QC$

Since not B, then $i = 10$, then:

$Inv \ \&\& \ \text{not } B = \{i = 10 \ \&\& \ k = B[0] + B[1] + \dots + B[9] \ \&\& \ |\{p : 0 \leq p < k \ \&\& \ A[p] = v\}| = B[v]$ for all v s.t. $0 \leq v < 10 \ \&\& \ |\{p : 0 \leq p < k \ \&\& \ A[p] = v\}| = 0$ for all v s.t. $10 \leq v \leq 9\}$

Thus we have $\{k = B[0] + B[1] + \dots + B[9] = A.length \ \&\& \ |\{p : 0 \leq p < k \ \&\& \ A[p] = v\}|$

$= B[v] \text{ for all } v \text{ s.t. } 0 \leq v \leq 9\}$

Since the process of sorting is appending blocks of value v where v is increasing from 0 to 9, thus A is nondecreasing, hence $\text{isSorted}(A)$ holds

Thus QC holds

2. For inner while loop (fixed i)

$PC2 = \{\text{fixed } i \text{ with } 0 \leq i \leq 9 \ \&\& \ j = 0 \ \&\& \ k = B[0] + B[1] + \dots + B[i-1] \ \&\& \ |\{p : 0 \leq p < k \ \&\& \ A[p] = v\}| = B[v] \text{ for all } v \text{ s.t. } 0 \leq v < i \ \&\& \ |\{p : 0 \leq p < k \ \&\& \ A[p] = v\}| = 0 \text{ for all } v \text{ s.t. } i \leq v \leq 9\}$

$Inv2 = \{0 \leq j \leq B[i] \ \&\& \ k = B[0] + B[1] + \dots + B[i-1] + j \ \&\& \ |\{p : B[0] + \dots + B[i-1] \leq p < k \ \&\& \ A[p] = i\}| = j \ \&\& \ |\{p : 0 \leq p < B[0] + \dots + B[i-1] \ \&\& \ A[p] = v\}| = B[v] \text{ for all } v \text{ s.t. } 0 \leq v < i\}$

$B2 = \{j < B[i]\}$

$QC2 = \{j = B[i] \ \&\& \ k = B[0] + B[1] + \dots + B[i] \ \&\& \ |\{p : 0 \leq p < k \ \&\& \ A[p] = i\}| = B[i] \ \&\& \ |\{p : 0 \leq p < k \ \&\& \ A[p] = v\}| = B[v] \text{ for all } v \text{ s.t. } 0 \leq v < i\}$

Then we prove correctness by loop invariant theorem

1. Initialization: $PC2 \Rightarrow Inv2$

Suppose $PC2$ holds, NTP $Inv2$

Since $j = 0$, then $0 \leq j \leq B[i]$

Since $k = B[0] + \dots + B[i-1]$, we get $k = B[0] + \dots + B[i-1] + j$

Since $k = B[0] + B[1] + \dots + B[i-1] \ \&\& \ |\{p : 0 \leq p < k \ \&\& \ A[p] = v\}| = B[v] \text{ for all } v \text{ s.t. } 0 \leq v < i$, then $|\{p : 0 \leq p < B[0] + B[1] + \dots + B[i-1] \ \&\& \ A[p] = v\}| = B[v] \text{ for all } v \text{ s.t. } 0 \leq v < i$

Since $|\{p : 0 \leq p < k \ \&\& \ A[p] = v\}| = 0 \text{ for all } v \text{ s.t. } i \leq v \leq 9$ and $j = 0$, then $|\{p : 0 \leq p < k \ \&\& \ A[p] = i\}| = 0$, then $|\{p : k \leq p < k \ \&\& \ A[p] = i\}| = 0$ since this set is empty, then $|\{p : B[0] + \dots + B[i-1] \leq p < k \ \&\& \ A[p] = i\}| = 0$ since $k = B[0] + \dots + B[i-1]$

Thus $Inv2$ holds

2. Maintenance: Assuming $Inv2$ & $B2$ holds, the body makes $Inv2$ true again

Since $B2$, then $j0 < B[i]$

Let $j = j0 + 1$; $k = k0 + 1$

Then $Inv2 \ \&\& \ B2 = \{0 \leq j0 < B[i] \ \&\& \ k0 = B[0] + B[1] + \dots + B[i-1] + j0 \ \&\& \ |\{p : B[0] + \dots + B[i-1] \leq p < k0 \ \&\& \ A[p] = i\}| = j0 \ \&\& \ |\{p : 0 \leq p < B[0] + \dots + B[i-1] \ \&\& \ A[p] = v\}| = B[v] \text{ for all } v \text{ s.t. } 0 \leq v < i\}$

Then we have $0 \leq j0 + 1 \leq B[i] \ \&\& \ k0 + 1 = B[0] + B[1] + \dots + B[i-1] + j0 + 1$

Then $0 \leq j \leq B[i] \ \&\& \ k = B[0] + B[1] + \dots + B[i-1] + j$

Since $|\{p : B[0] + \dots + B[i-1] \leq p < k0 \ \&\& \ A[p] = i\}| = j0$, then $|\{p : B[0] + \dots + B[i-1] \leq p < k0 + 1 \ \&\& \ A[p] = i\}| = j0 + |\{p : A[k0 + 1] = i\}|$

Since we know $A[k0 + 1] = A[k] = i$ happens in the while loop body, then $|\{p : B[0] + \dots + B[i-1] \leq p < k0 + 1 \ \&\& \ A[p] = i\}| = j0 + 1 = j$

Since $|\{p : 0 \leq p < B[0] + \dots + B[i-1] \ \&\& \ A[p] = v\}| = B[v] \text{ for all } v \text{ s.t. } 0 \leq v < i$ and in the loop body we have $A[k] = i$, this doesn't change the value of $B[0]$ to $B[i-1]$ because $A[k] = i$ will only effect $B[i]$

Thus this invariant holds again

3. Termination: $Inv2 \ \&\& \ \text{not } B2 \Rightarrow QC2$

Since $Inv2$ we have $j \leq B[i]$, and since $\text{not } B2$ we have $j \geq B[i]$, thus $j = B[i]$

Also since $k = B[0] + B[1] + \dots + B[i-1] + j$, then $k = B[0] + B[1] + \dots + B[i]$

In order to prove $|\{p : 0 \leq p < k \ \&\& \ A[p] = i\}| = B[i]$

Let $s = B[0] + \dots + B[i-1]$, then $|\{p : 0 \leq p < k \ \&\& \ A[p] = i\}| = |\{p : 0 \leq p < s \ \&\& \ A[p] = i\}| + |\{p : s \leq p < k \ \&\& \ A[p] = i\}|$

$A[p] = i \} + |\{p : s \leq p < k \ \&\& \ A[p] = i \}|$

Since the outer-loop precondition of the inner loop: $|\{p : 0 \leq p < s \ \&\& \ A[p] = i \}| = 0$, and from the inner-loop invariant: $|\{p : s \leq p < k \ \&\& \ A[p] = i \}| = j = B[i]$

Then the total is $0 + B[i] = B[i]$

In order to prove $|\{p : 0 \leq p < k \ \&\& \ A[p] = v \}| = B[v]$ for all v s.t. $0 \leq v < i$

$|\{p : 0 \leq p < k \ \&\& \ A[p] = v \}| = |\{p : 0 \leq p < s \ \&\& \ A[p] = v \}| + |\{p : s \leq p < k \ \&\& \ A[p] = v \}|$

Since Inv2: $|\{p : 0 \leq p < s \ \&\& \ A[p] = v \}| = B[v]$ for $0 \leq v < i$ and since the inner loop's has $A[k] = i$ and it fills the indices $p : s \leq p < k$ before terminating, then every $A[p] = i$, then for any $v < i$, no $p : s \leq p < k$ satisfies $A[p] = v$

Thus the set $\{p : s \leq p < k \ \&\& \ A[p] = v \}$ is empty, thus $|\{p : s \leq p < k \ \&\& \ A[p] = v \}| = 0$

Thus the sum is $B[v] + 0 = B[v]$

Thus Qc holds

(d) Implement the algorithm in Java

Just follow the pseudo code

(e) Supply meaningful unit tests

There are two precondition, so I create two negative test to test if the precondition works well

Then I create a positive test to test if this algorithm can sort the array well

Problem 03: Palindrome Collector

(a) Document pre-/post-conditions in clear English without revealing any internal representation

1. `addFirst`

Precondition:

1. input char should be lower case and be in [a,z]
2. collector should not be full

Postcondition:

1. the length of new collector increase by 1 after adding
2. the first element of new collector becomes ch
3. from the second element to the last element of the new collector equals the elements in original collector in order

2. `addLast`

Precondition:

1. input char should be lower case and be in [a,z]
2. collector should not be full

Postcondition:

1. the length of new collector increase by 1 after adding
2. the last element of new collector becomes ch
3. from the first element to the previous one of the last element of the new collector equals the elements in original collector in order

3. `removeFirst`

Precondition:

1. collector should not be empty

Postcondition:

1. the length of new collector decrease by 1 after removing
2. the first element of new collector becomes the second element of original collector
3. from the second element to the last element of the original collector equals the elements in new collector in order
4. return the character that is the first element of original collector

4. `removeLast`

Precondition:

1. collector should not be empty

Postcondition:

1. the length of new collector decrease by 1 after removing
2. the last element of new collector becomes the previous element of the last element of original collector
3. from the first element to the previous element of the last element of the original collector equals the elements in new collector in order
4. return the character that is the last element of original collector

5. `isPalindrome`

Precondition:

None

Postcondition:

1. return true iff the collector reads the same forward and backward
2. the collector is not modified

6. `isEmpty`

Precondition:

None

Postcondition:

1. return true iff the size of collector is 0
2. the collector is not modified

7. `size`

Precondition:

None

Postcondition:

1. return the number of characters in the collector
2. the collector is not modified

(b) Declare all fields needed to guarantee $O(1)$ time for every operation except for `isPalindrome`

1. A char array `char[] collector` that represents the collector, since the collector is a fixed-size circular char array
2. Capacity of collector `int capacity` that fixes the max number of characters that the collector can have
3. `DEFAULT_CAPACITY` `final int DEFAULT_CAPACITY` that is a constant capacity and is used when we construct the collector without passing parameter: capacity
4. The head of the collector `int head` that indicate the first element in the collector, since the array is circular and we will do insertion and removal, then head will move, then we need its position
5. The tail of the collector `int tail` that indicate the last element in the collector, since the array is circular and we will do

insertion and removal, then tail will move, then we need its position

6. The size of the collector `int size` that tell us the currently used size of collector, since we need to know if the collector is full

If we have those fields, then we will have a collector that is similar to double-ended queue

1. `addFirst`

This method is $O(1)$ since we know the position of first element: `head`, then we just move `head` forward one step and add element in that position and increase the `size`

2. `addLast`

This method is $O(1)$ since we know the position of last element: `tail`, then we just move `tail` backward one step and add element in that position and increase the `size`

3. `'removeFirst'`

This method is $O(1)$ since we know the position of last element: `head`, then we can read the first element and return it. To remove it, we move `head` backward one step and decrease the `size`

4. `'removeLast'`

This method is $O(1)$ since we know the position of last element: `tail`, then we can read the last element and return it. To remove it, we move `tail` forward one step and decrease the `size`

5. `'isEmpty'`

This method is $O(1)$ since we have the field `size`, then just check if the `size` is 0

6. `'size()` This method is $O(1)$ since we directly return the field `size`'

(c) Provide a representation invariant expressed in English and describe an abstract function that explains, in terms of the ADT, what the concrete state means

Representation invariant

1. Capacity is positive and not null
2. The length of collector equals capacity
3. Size is bounded by 0(included) and capacity(included)
4. Head and tail are valid: ranging from 0(included) to capacity(excluded)
5. Head, tail and size satisfy the property of circular array
6. Every character in collector is lowercase letter from 'a' to 'z'

$RI(R) = capacity > 0 \wedge collector.length = capacity \wedge 0 \leq size \leq capacity \wedge 0 \leq head, tail < capacity \wedge tail = (head + size) \bmod capacity \wedge \forall i < size: 'a' \leq collector[(head+i) \bmod capacity] \leq 'z'$

Abstraction function

- Abstract values (A): finite sequences of lowercase characters `['a'..'z']`
- Representation values (R): `R = <collector : char[capacity], capacity : int, head : int, tail : int, size : int>`
- Abstraction Function

Let `AF : R  $\Rightarrow$  A` s.t.

$(AF(R) = L \iff |L| = R.size \wedge \forall k \in \{1, \dots, R.size\} : L[k] = R.collector[(R.head + k - 1) \bmod R.capacity])$

In English: the abstract list `L` is the `size` elements starting at `head` in `collector`, max size limited by `capacity`